



Suez Canal University

Faculty of Computers and Informatics

Computer Science Department



Operating Systems

Practical Training Book

A Hands-On Guide to Linux System Administration

Red Hat Enterprise Linux (RHEL) Focus

Author: Mostafa Elkhoully

Assistant Lecturer

Computer Science Department

Faculty of Computers and Informatics

Suez Canal University

Beginner to Intermediate Level

© 2026 Suez Canal University. All Rights Reserved.

Preface

This practical training book is designed to provide university students with a comprehensive, hands-on introduction to Operating Systems, with a strong focus on Red Hat Enterprise Linux (RHEL). The book covers essential concepts and practical skills needed for effective Linux system administration.

The material is organised into ten chapters, progressing from fundamental concepts to intermediate system administration tasks. Each chapter includes clear learning objectives, theoretical explanations, detailed command references with examples, step-by-step practical labs, review questions, and quizzes.

By the end of this book, you will have gained the foundational knowledge and practical experience necessary to work confidently with Linux systems in both academic and professional environments.

Target Audience

- Third-level students in the Computer Science Department
- Second-level students in the Software Engineering Department
- Second-level students in the Artificial Intelligence Department
- University students with beginners to intermediate knowledge of Linux and Red Hat
- System administration trainees preparing for RHCSA certification

Prerequisites

- Basic computer literacy
- Access to a Linux system (RHEL, CentOS, or Fedora recommended)
- Willingness to practise commands in a terminal

Table of Contents

- Chapter 1:** Introduction to Linux and Open Source
- Chapter 2:** Linux Installation
- Chapter 3:** Linux Components and Terminal Basics
- Chapter 4:** Linux File System and Navigation
- Chapter 5:** File and Directory Management
- Chapter 6:** User and Group Administration
- Chapter 7:** File Permissions and Ownership
- Chapter 8:** Vi/Vim Text Editor
- Chapter 9:** Environment Variables and Shell Configuration
- Chapter 10:** System Administration Basics

Chapter 1: Introduction to Linux and Open Source

Learning Objectives

After completing this chapter, you will be able to:

- Understand what Free and Open Source Software (FOSS) means
 - Identify common open source licenses
 - Describe the history of Unix and Linux
 - List major Linux distributions
 - Explain why organizations choose Linux and Red Hat
-

1.1 What is Free and Open Source Software (FOSS)?

Free/Open Source Software (FOSS) provides many freedoms, including the ability to:

- **View the source code** used to compile programs
- **Make modifications** to the software
- **Distribute these modifications** to others

Note: "Free" in FOSS refers to freedom, not price. Think "free speech," not "free beer."

1.2 Open Source Licenses

An open-source license allows the source code, blueprint, or design to be used, modified, and/or shared under defined terms and conditions.

Common Open Source Licenses

License	Description
GPL	GNU General Public License - Copyleft license requiring derivatives to be open source
LGPL	Lesser GPL - Allows linking with proprietary software
Apache	Permissive license with patent protection
BSD	Permissive license with minimal restrictions
Mozilla	Hybrid license used by Firefox

Most FOSS is covered under the **GNU General Public License (GPL)**, which is the most common public license.

1.3 Linux History

The Origins of Unix

Year	Event
1963-1969	Multics developed by MIT, General Electric, and Bell Labs
1969	First version of Unix created at Bell Labs
1983	Richard Stallman started the GNU project

Unix Flavors

Different companies created their own Unix variants:

Company	Unix Variant
IBM	AIX
Hewlett-Packard	HP/UX
Sun Microsystems	Solaris
Silicon Graphics	IRIX

These systems:

- Operate in the same manner
- Offer the same standard utilities and commands

The Birth of Linux

Year	Event
1983	Richard Stallman started GNU project with goal of creating a free Unix-like OS
1991	Linus Torvalds created the Linux kernel
1992	Linux and GNU developers integrated to create a fully functional free OS

Key Organizations

- **Free Software Foundation (FSF)** - Founded by Richard Stallman
 - **GNU Project** - Created essential software components
 - **Linux Foundation** - Supports Linux development
-

1.4 Linux Distributions

A Linux distribution (distro) is a complete operating system built around the Linux kernel.

Major Linux Distribution Families

+-----+
Debian-based
+-----+
-- Ubuntu
-- Linux Mint
-- Kubuntu
+-----+
Red Hat-based
+-----+
-- RHEL
-- CentOS
-- Fedora
+-----+
Arch-based
+-----+
-- Manjaro
-- BlackArch

Popular Distributions

Distribution	Use Case
Red Hat Enterprise Linux	Enterprise servers
Fedora	Desktop and development
Ubuntu	Desktop and cloud
CentOS	Community enterprise
Debian	Servers and stability

Tip: Visit distrowatch.com to explore Linux distributions.

1.5 Why Choose Linux?

Linux offers many advantages:

1. **Open Source** - Released under GNU GPL
2. **Security** - More secure and virus-resistant
3. **Stability** - Excellent for servers
4. **Flexibility** - Highly customizable
5. **Cost** - Free to use
6. **Community** - Strong community support
7. **Programming** - Perfect for developers

Linux Market Dominance

- Dominates the server market
- Powers most Internet infrastructure
- Used by ISPs, e-commerce sites, and commercial applications
- Runs on supercomputers, smartphones (Android), and IoT devices

1.6 Why Red Hat Enterprise Linux?

Red Hat was founded on March 26, 1993, and Red Hat Linux first appeared in 1994.

Advantages of RHEL

- More than **90% of Fortune Global 500** companies use Red Hat products
- The most demanding applications run better on RHEL
- **Scales well** and is more reliable
- **Secure** by design
- Red Hat partnership with hardware vendors
- Professional training and support available
- Industry-recognized certifications (RHCSA, RHCE)

1.7 Linux Rules to Remember

Rule 1: Everything in Linux is a file

Rule 2: Linux is case sensitive (File.txt ≠ file.txt)

Practical Lab 1: Exploring Linux

Lab Objectives

- Access a Linux terminal
- Identify the Linux distribution
- View system information

Lab Steps

Step 1: Open a terminal

Step 2: Check the operating system name

```
$ uname  
Linux
```

Step 3: View detailed system information

```
$ uname -a  
Linux hostname 4.18.0-305.el8.x86_64 #1 SMP x86_64 GNU/Linux
```

Step 4: Check the distribution version

```
$ cat /etc/redhat-release  
Red Hat Enterprise Linux release 8.4 (Ootpa)
```

Step 5: View hostname

```
$ uname -n  
hostname
```

Expected Results

- You should see "Linux" as the OS name
- Detailed information includes kernel version
- Distribution file shows RHEL version

Review Questions

8. What does FOSS stand for?
 9. What is the most common open source license?
 10. Who created the Linux kernel and when?
 11. Name three Linux distributions based on Red Hat.
 12. What percentage of Fortune 500 companies use Red Hat products?
 13. What are the two fundamental rules of Linux?
-

Hands-on Exercises

14. Research and list 5 devices that run Linux
 15. Visit distrowatch.com and find the top 10 distributions
 16. Compare the GPL and BSD licenses
 17. Find out which Linux distribution your university uses
-

Chapter Quiz

1. FOSS stands for:

- a) Free Operating System Software
- b) Free/Open Source Software ✓ Correct
- c) Fast Open Source System
- d) File Operating System Standard

2. The Linux kernel was created by:

- a) Richard Stallman
- b) Bill Gates
- c) Linus Torvalds ✓ Correct
- d) Steve Jobs

3. Which year was Linux kernel created?

- a) 1983
- b) 1989
- c) 1991 ✓ Correct
- d) 1995

4. Which is NOT a Red Hat-based distribution?

- a) Fedora
- b) CentOS
- c) Ubuntu ✓ Correct
- d) RHEL

5. In Linux, "File.txt" and "file.txt" are:

- a) The same file
 - b) Different files ✓ Correct
 - c) Invalid names
 - d) System files
-

Chapter 2: Linux Installation

Learning Objectives

After completing this chapter, you will be able to:

- Understand different Linux installation methods
 - Identify hardware requirements for RHEL
 - Perform a complete RHEL installation
 - Configure post-installation settings
-

2.1 Types of Installation

Installation Methods

Method	Description
Graphical Installation	GUI-based installer (Anaconda)
Text-Based Installation	For systems without graphics capability
Kickstart (Automated)	Unattended installation using configuration file
Remote Installation	Network-based installation (PXE boot)

2.2 Hardware Requirements

Minimum Requirements for RHEL 8

Component	Minimum	Recommended
CPU	1 core	2+ cores
RAM	1.5 GB	4+ GB
Disk Space	10 GB	20+ GB
Network	Optional	Recommended

2.3 Pre-Installation Checklist

Before installing, gather this information:

- ☐ Installation media (USB/DVD/ISO)
 - ☐ Network configuration (IP, Gateway, DNS)
 - ☐ Hostname
 - ☐ Partition plan
 - ☐ Root password
 - ☐ User accounts to create
 - ☐ Time zone
-

2.4 Step-by-Step Installation Guide

Step 1: Boot from Installation Media

Insert USB or mount ISO and boot the system.

Step 2: Select "Install Red Hat Enterprise Linux"

Choose installation option from the boot menu.

Step 3: Language Selection

Select your preferred language for the installation.

Step 4: Installation Summary Screen

Configure the following sections:

Localization

- **Keyboard:** Select keyboard layout
- **Language Support:** Add additional languages
- **Time & Date:** Set timezone and NTP

Software

- **Installation Source:** Local media or network
- **Software Selection:** Choose base environment
 - Server with GUI
 - Server
 - Minimal Install
 - Workstation

System

- **Installation Destination:** Configure disk partitions

Step 5: Disk Partitioning

Recommended Partition Scheme

Mount Point	Size	Purpose
/boot	1-2 GB	Boot loader and kernel
/	10-20 GB	Root filesystem
/home	Remaining	User data
swap	2x RAM (up to 8GB)	Virtual memory

Step 6: Network Configuration

- Set hostname
- Configure network interface
- Enable automatic connection

Step 7: Begin Installation

Click "Begin Installation" to start.

Step 8: User Settings (During Installation)

- Set **Root Password**
- Create **User Account**

Step 9: Complete Installation

- Reboot when installation completes
- Accept license agreement on first boot

2.5 Post-Installation Tasks

After installation, perform these tasks:

```
# Update the system
$ sudo yum update -y

# Check system information
$ cat /etc/redhat-release

# Verify network connectivity
$ ping -c 4 google.com

# Set hostname permanently
$ sudo hostnamectl set-hostname myserver
```

Practical Lab 2: Installing RHEL in VirtualBox

Lab Objectives

- Create a virtual machine
- Install RHEL from ISO
- Configure basic settings

Prerequisites

- VirtualBox installed
- RHEL ISO image downloaded

Lab Steps

Step 1: Create New Virtual Machine

- Name: RHEL-Lab
- Type: Linux
- Version: Red Hat (64-bit)
- Memory: 2048 MB
- Create virtual hard disk: 20 GB

Step 2: Mount ISO

- Settings → Storage → Controller: IDE
- Add optical drive → Choose ISO

Step 3: Start VM and Install

- Follow installation steps above
- Use automatic partitioning

Step 4: Verify Installation

```
$ uname -a
$ cat /etc/redhat-release
$ df -h
```

Expected Results

- VM boots into RHEL
- System information displays correctly
- Disk partitions are created

Review Questions

18. What is the name of the RHEL graphical installer?
 19. What is Kickstart installation used for?
 20. What is the minimum RAM required for RHEL 8?
 21. What is the purpose of the swap partition?
 22. Which command updates all packages on RHEL?
-

Chapter Quiz

1. The RHEL graphical installer is called:

- a) YUM
- b) Anaconda ✓ Correct
- c) RPM
- d) Grub

2. Kickstart installation is:

- a) Faster than graphical
- b) Automated/unattended ✓ Correct
- c) Text-only
- d) Network-based only

3. The recommended size for /boot partition is:

- a) 512 MB
- b) 1-2 GB ✓ Correct
- c) 10 GB
- d) 50 GB

4. Which command sets the hostname permanently?

- a) hostname
 - b) hostnamectl ✓ Correct
 - c) sethostname
 - d) sethost
-

Chapter 3: Linux Components and Terminal Basics

Learning Objectives

After completing this chapter, you will be able to:

- Describe the Linux kernel and its functions
 - Understand the role of the shell
 - Use the terminal effectively
 - Execute basic commands with proper syntax
 - Navigate between virtual consoles
-

3.1 The Linux Kernel

The kernel is the **core of the operating system**.

Kernel Functions

- Manages hardware devices through device drivers
- Controls process scheduling
- Handles memory management
- Provides system calls for programs

Key Facts

- Loads into RAM when machine boots
- Stays resident in RAM until power off
- Current kernel info: `uname -r`

```
$ uname -r  
4.18.0-305.el8.x86_64
```

3.2 The Shell

The shell provides an interface for users to communicate with the kernel.

Shell Functions

- Parses commands entered by users
- Translates commands into executable segments
- Provides programming capabilities (scripting)

Common Shells

Shell	Name	Path
sh	Bourne Shell	/bin/sh
bash	Bourne Again Shell	/bin/bash
csh	C Shell	/bin/csh
ksh	Korn Shell	/bin/ksh
zsh	Z Shell	/bin/zsh

Note: bash is the most commonly used shell on Linux and is the default in RHEL.

Check your current shell:

```
$ echo $SHELL
/bin/bash
```

3.3 The Terminal

The terminal gives the shell a place to:

- Accept typed commands
- Display command results

Types of Terminals

- **Terminal Emulator** - GUI application (GNOME Terminal, Konsole)
- **Virtual Console (TTY)** - Text-only interface

3.4 Command Syntax

All Linux commands follow this structure:

```
command [options] [arguments]
```

Component	Description	Example
command	The program to run	ls
options	Modify behavior (start with -)	-l, -a
arguments	Files or data to operate on	/home

Examples

```
# Command only
$ date
Mon Feb 9 12:30:45 EET 2026
```

```
# Command with option
$ ls -l
total 4
drwxr-xr-x. 2 root root 4096 Feb 9 10:00 Documents

# Command with argument
$ ls /home
user1 user2

# Command with option and argument
$ ls -l /home
drwx----- 3 user1 user1 4096 Feb 9 10:00 user1

# Multiple commands (separated by semicolon)
$ date; whoami
Mon Feb 9 12:30:45 EET 2026
root
```

3.5 Virtual Consoles (TTY)

Linux provides multiple text-based terminals called TTY (Teletypewriter).

Switching Consoles

Shortcut	Description
Ctrl+Alt+F1	Return to GUI (GNOME)
Ctrl+Alt+F2	TTY 2
Ctrl+Alt+F3	TTY 3
Ctrl+Alt+F4	TTY 4
Ctrl+Alt+F5	TTY 5
Ctrl+Alt+F6	TTY 6

From TTY, use Alt+Fn to switch between consoles.

Check Current TTY

```
$ tty
/dev/pts/0

$ chvt 3    # Change to virtual terminal 3
```

Use Case: TTY is essential when GUI fails or for server administration.

3.6 Essential Keyboard Shortcuts

Shortcut	Action
Ctrl+C	Stop/cancel current command
Ctrl+D	Exit/logout (end of input)
Ctrl+L	Clear terminal screen
Ctrl+Z	Pause current process
Ctrl+A	Move cursor to beginning of line
Ctrl+E	Move cursor to end of line
Ctrl+K	Delete from cursor to end of line
Ctrl+U	Delete from cursor to beginning
Ctrl+S	Lock screen (pause output)
Ctrl+Q	Unlock screen
Tab	Auto-complete commands/files
↑ / ↓	Navigate command history

3.7 Interrupting Command Execution

Stop a Running Command

```
$ ping google.com
PING google.com...
^C          # Press Ctrl+C to stop
```

Terminate Input

```
$ cat
Hello
World
^D          # Press Ctrl+D to end input
```

3.8 Basic Information Commands

System Information

Command	Purpose	Example Output
uname	Print system name	Linux
uname -a	All system info	Linux host 4.18.0...
uname -n	Hostname	myserver
uname -r	Kernel release	4.18.0-305.el8
hostname	Display hostname	myserver

Date and Time

Command	Purpose	Example Output
date	Current date/time	Mon Feb 9 12:30:45
cal	Current month calendar	(calendar display)
cal 2026	Year calendar	(full year)
cal 3 2026	Specific month	March 2026

User Information

Command	Purpose	Example Output
whoami	Current username	root
id	User and group IDs	uid=0(root) gid=0(root)
who	Logged in users	root pts/0 Feb 9
w	Detailed user activity	(detailed list)
users	List of users	root user1

```
$ id
uid=1000(mostafa) gid=1000(mostafa) groups=1000(mostafa),10(wheel)

$ who
mostafa pts/0      2026-02-09 12:30 (:0)

$ w
12:35  up 2:30,  1 user,  load average: 0.00, 0.01, 0.05
USER    TTY      FROM          LOGIN@   IDLE   WHAT
mostafa pts/0    :0            12:30    0.00s  w
```

Practical Lab 3: Terminal Navigation

Lab Objectives

- Access different terminals
- Execute basic commands
- Use keyboard shortcuts

Lab Steps

Step 1: Open a terminal emulator in GUI

Step 2: Execute information commands

```
$ uname -a
$ date
$ whoami
$ id
```

Step 3: Practice keyboard shortcuts

- Type a long command, use Ctrl+A and Ctrl+E
- Type `sleep 100`, then Ctrl+C to stop
- Type `clear` or use Ctrl+L

Step 4: Switch to TTY (Ctrl+Alt+F2)

- Login with your credentials
- Run `tty` to confirm
- Return to GUI: Ctrl+Alt+F1

Step 5: Use multiple commands

```
$ date; uname; whoami
```

Review Questions

- 23. What is the kernel's primary function?
- 24. What is the default shell in RHEL?
- 25. How do you switch to TTY 3?
- 26. What does Ctrl+C do?
- 27. What command shows your user ID?

Chapter Quiz

1. The core component of Linux is:

- a) Shell
- b) Terminal
- c) Kernel ✓ Correct
- d) GNOME

2. The default shell in RHEL is:

- a) csh
- b) ksh
- c) bash ✓ Correct
- d) zsh

3. To cancel a running command, press:

- a) Ctrl+D
- b) Ctrl+C ✓ Correct
- c) Ctrl+Z
- d) Ctrl+X

4. Virtual consoles are accessed with:

- a) Alt+Tab
- b) Ctrl+Alt+Fn ✓ Correct
- c) Shift+Fn
- d) Super+Fn

Chapter 4: Linux File System and Navigation

Learning Objectives

After completing this chapter, you will be able to:

- Understand the Linux file system hierarchy
- Identify the purpose of important directories
- Navigate using absolute and relative paths
- Use navigation commands effectively

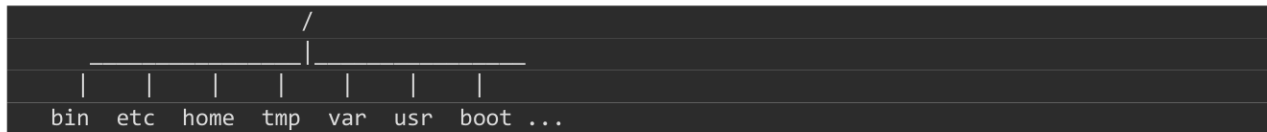
4.1 Linux File System Hierarchy

Think of the file system as a building:

- **File system** = Building
- **Directory** = Room
- **File** = Desk
- **Current directory** = Room you are in

The Root Directory (/)

Everything in Linux starts from the root directory /. This is the top of the file system tree.



4.2 Important Directories

Directory	Purpose	Notes
/	Root file system	Top-level directory
/home	Normal users' home directories	/home/mostafa, /home/ahmed
/root	Root user's home directory	Only for superuser
/etc	Configuration files	System and service configs
/dev	Device files	Hardware represented as files
/tmp	Temporary files	Deleted after reboot
/proc	Process information	Kernel runtime data (read-only)
/sys	System hardware info	Kernel hardware data
/var	Variable data	Logs, databases, emails
/usr	User shared files	Programs, libraries, docs
/bin	Essential user commands	ls, cp, cat, etc.
/sbin	System admin commands	Root/admin commands

/opt	Optional software	Third-party applications
/boot	Boot files	Kernel and bootloader
/lib	System libraries	32-bit libraries
/lib64	64-bit libraries	64-bit system libraries
/media	Removable media	USB, CD-ROM mount points
/mnt	Temporary mounts	Manual mount point
/srv	Service data	Web server data, etc.

4.3 Pathnames

Absolute Pathname

- Starts from root (/)
- Complete path to file/directory
- Always works regardless of current location

```
/home/mostafa/Documents/report.txt
/etc/passwd
/var/log/messages
```

Relative Pathname

- Relative to current directory
- Does not start with /
- Uses . (current) and .. (parent)

```
Documents/report.txt # From /home/mostafa
../ahmed/             # Parent's ahmed directory
./script.sh           # Current directory
```

Special Path Symbols

Symbol	Meaning	Example
/	Root directory	cd /
.	Current directory	./script.sh
..	Parent directory	cd ..
~	Home directory	cd ~
~user	User's home	cd ~mostafa
-	Previous directory	cd -

4.4 Navigation Commands

pwd - Print Working Directory

Shows your current location in the file system.

Syntax: pwd

```
$ pwd
/home/mostafa
```

cd - Change Directory

Move between directories.

Syntax: cd [directory]

Command	Action
cd /var/log	Go to absolute path
cd Documents	Go to relative path
cd ..	Go to parent directory
cd or cd ~	Go to home directory
cd -	Go to previous directory
cd ~mostafa	Go to mostafa's home

```
$ pwd
/home/mostafa
```

```
$ cd /var/log
$ pwd
/var/log
```

```
$ cd -
/home/mostafa
```

```
$ cd ..
$ pwd
/home
```

ls - List Directory Contents

View files and directories.

Syntax: ls [options] [directory]

Option	Description
-l	Long listing (detailed)
-a	Show all (including hidden)
-h	Human-readable sizes
-R	Recursive listing
-r	Reverse order
-t	Sort by time
-F	Show file type indicators
-d	List directory itself

```
$ ls
Documents Downloads Pictures

$ ls -l
total 12
drwxr-xr-x. 2 mostafa mostafa 4096 Feb 9 10:00 Documents
drwxr-xr-x. 2 mostafa mostafa 4096 Feb 9 10:00 Downloads
drwxr-xr-x. 2 mostafa mostafa 4096 Feb 9 10:00 Pictures

$ ls -la
total 20
drwx-----. 5 mostafa mostafa 4096 Feb 9 10:00 .
drwxr-xr-x. 3 root root 4096 Feb 9 09:00 ..
-rw-----. 1 mostafa mostafa 100 Feb 9 10:00 .bash_history
drwxr-xr-x. 2 mostafa mostafa 4096 Feb 9 10:00 Documents

$ ls -F
Documents/ Downloads/ script.sh* link@
```

File Type Indicators (ls -F)

Indicator	Type
/	Directory
*	Executable file
@	Symbolic link
(none)	Regular file

File Colors (default)

Color	Type
Blue	Directory
White	Normal file
Green	Executable
Cyan	Symbolic link
Red	Compressed file

4.5 Hidden Files

Files starting with **.** are hidden.

```
$ ls
Documents

$ ls -a
. .. .bashrc .profile Documents
```

- **.** = Current directory
- **..** = Parent directory
- **.bashrc** = Hidden configuration file

Practical Lab 4: File System Navigation

Lab Objectives

- Navigate the file system
- Use absolute and relative paths
- List directory contents

Lab Steps

Step 1: Check current location

```
$ pwd  
/home/student
```

Step 2: Navigate using absolute paths

```
$ cd /etc  
$ pwd  
/etc  
$ ls | head -5
```

Step 3: Navigate using relative paths

```
$ cd ~  
$ cd Documents  
$ pwd
```

Step 4: Use special symbols

```
$ cd /var/log  
$ cd ..  
$ pwd  
/var  
  
$ cd -  
/var/log  
  
$ cd ~  
$ pwd  
/home/student
```

Step 5: List with different options

```
$ ls -la /etc | head -10  
$ ls -lh /var/log  
$ ls -R ~/Documents
```

Step 6: Explore important directories

```
$ ls /bin | head -5
$ ls /etc | head -5
$ cat /etc/hostname
```

Verification

- Confirm you can navigate between directories
- Verify ls shows different output with options

Review Questions

- 28. What is the root directory in Linux?
- 29. Where are user home directories located?
- 30. What does the . symbol represent?
- 31. What command shows your current directory?
- 32. How do you go to the previous directory?

Chapter Quiz

1. The root directory is represented by:

- a) /root
- b) / **✓ Correct**
- c) ~
- d) ..

2. Hidden files in Linux start with:

- a) #
- b) _
- c) . **✓ Correct**
- d) ~

3. To go to your home directory, type:

- a) cd /home
- b) cd ~ **✓ Correct**
- c) cd ..
- d) cd root

4. ls -a shows:

- a) All users
- b) All including hidden files **✓ Correct**
- c) Archives only
- d) Admin files

Chapter 5: File and Directory Management

Learning Objectives

After completing this chapter, you will be able to:

- Create files and directories
- View file contents using various commands
- Copy, move, and rename files
- Remove files and directories
- Use wildcards for file operations

5.1 Creating Files

touch - Create Empty Files

Syntax: touch [options] filename(s)

Usage	Description
touch file1	Create empty file
touch file1 file2 file3	Create multiple files
touch /tmp/file1	Create file in specific path
On existing file	Updates timestamp

```
$ touch myfile.txt
$ ls -l myfile.txt
-rw-r--r--. 1 mostafa mostafa 0 Feb 9 12:00 myfile.txt

$ touch file1 file2 file3
$ ls
file1 file2 file3 myfile.txt
```

5.2 Creating Directories

mkdir - Make Directory

Syntax: mkdir [options] directory(s)

Option	Description
(none)	Create directory
-p	Create parent directories if needed

```
$ mkdir Documents

$ mkdir dir1 dir2 dir3

$ mkdir /tmp/mydir

$ mkdir -p projects/year2026/reports
# Creates all parent directories
```

5.3 Viewing File Contents

cat - Concatenate and Display

Syntax: cat [options] filename

```
$ cat /etc/hostname
myserver

$ cat file1 file2    # View multiple files
```

more - Page Through File

Syntax: more filename

Key	Action
Space	Forward one screen
Enter	Forward one line
b	Back one screen
/text	Search for text
n	Next search match
q	Quit

less - Better Paging (Recommended)

Syntax: less filename

Same keys as **more**, plus:

- Arrow keys for navigation
- Can scroll backwards

head - View Beginning

Syntax: head [-n count] filename

```
$ head /etc/passwd    # First 10 lines
$ head -n 5 /etc/passwd # First 5 lines
$ head -5 /etc/passwd  # Same as above
```

tail - View End

Syntax: tail [-n count] filename

```
$ tail /var/log/messages      # Last 10 lines
$ tail -n 20 /var/log/messages # Last 20 lines
$ tail -f /var/log/messages    # Follow (live updates)
```

5.4 Copying Files and Directories

cp - Copy

Syntax: cp [options] source destination

Option	Description
-i	Interactive (prompt before overwrite)
-r	Recursive (for directories)
-v	Verbose output
-p	Preserve attributes

```
# Copy file to new name
$ cp file1 file1_backup

# Copy file to directory
$ cp file1 /tmp/

# Copy file to directory with new name
$ cp file1 /tmp/newfile

# Copy directory (must use -r)
$ cp -r Documents /tmp/

# Copy multiple files to directory
$ cp file1 file2 file3 /tmp/
```

5.5 Moving and Renaming

mv - Move/Rename

Syntax: mv [options] source destination

Option	Description
-i	Interactive (prompt before overwrite)
-v	Verbose output

```
# Rename file
$ mv oldname.txt newname.txt

# Move file to directory
$ mv file1 /tmp/

# Move with new name
$ mv file1 /tmp/newfile

# Move directory (no -r needed)
$ mv Documents /backup/

# Move multiple files
$ mv file1 file2 file3 /destination/
```

5.6 Removing Files and Directories

rm - Remove Files

Syntax: rm [options] file(s)

Option	Description
-i	Interactive (confirm each)
-r	Recursive (for directories)
-f	Force (no prompts)
-v	Verbose output

```
# Remove file
$ rm file1

# Remove multiple files
$ rm file1 file2 file3

# Remove with confirmation
$ rm -i important.txt
rm: remove regular file 'important.txt'? y
```

rmdir - Remove Empty Directory

Syntax: rmdir directory

```
$ rmdir emptydir
# Only works if directory is empty
```

rm -r - Remove Directory with Contents

```
# Remove directory and contents
$ rm -r mydir/
```

```
# Force remove (DANGEROUS!)  
$ rm -rf mydir/
```

WARNING: rm -rf is extremely dangerous. Always verify the path before using!

5.7 File Globbing (Wildcards)

Wildcards allow pattern matching for file names.

Asterisk (*) - Zero or More Characters

```
$ ls *.txt          # All .txt files  
file1.txt  file2.txt  notes.txt  
  
$ ls file*          # Files starting with "file"  
file1  file2  file3  file.txt  
  
$ ls *report*       # Contains "report"  
annual_report.pdf  report.doc
```

Question Mark (?) - Single Character

```
$ ls file?          # file + one character  
file1  file2  file3  
  
$ ls file??         # file + two characters  
file10  file99  
  
$ ls ???            # Exactly 3 characters  
abc  dir  tmp
```

Square Brackets ([]) - Character Range

```
$ ls file[123]       # file1, file2, or file3  
file1  file2  file3  
  
$ ls file[1-5]       # file1 through file5  
file1  file2  file3  file4  file5  
  
$ ls [a-z]*          # Starts with lowercase  
abc.txt  document.pdf  
  
$ ls [A-Za-z]*       # Starts with any letter
```

Globbering Examples

```
$ ls -a
.  ..  .profile  abm  bam  bat  battle  project

$ ls b*
bam  bat  battle

$ ls *m
abm  bam

$ ls ???
abm  bam  bat

$ ls ?a?
bam  bat

$ ls [ab]*
abm  bam  bat  battle
```

Practical Lab 5: File Operations

Lab Objectives

- Create and manage files and directories
- Copy, move, and rename items
- Practice using wildcards

Lab Steps

Step 1: Create working directory

```
$ cd ~
$ mkdir lab5
$ cd lab5
```

Step 2: Create files

```
$ touch file1.txt file2.txt file3.txt
$ touch report1.doc report2.doc
$ ls
```

Step 3: Create directories

```
$ mkdir backup documents
$ mkdir -p projects/2026/reports
$ ls -R
```

Step 4: Copy operations

```
$ cp file1.txt backup/  
$ cp -r projects /tmp/  
$ ls backup/
```

Step 5: Move and rename

```
$ mv file2.txt documents/  
$ mv file3.txt renamed.txt  
$ ls
```

Step 6: Practice wildcards

```
$ ls *.txt  
$ ls file*  
$ ls report?.doc  
$ cp *.doc documents/
```

Step 7: Cleanup

```
$ rm renamed.txt  
$ rm -r backup/  
$ cd ~  
$ rm -r lab5/
```

Review Questions

33. What command creates an empty file?
34. What option is needed to copy directories?
35. How do you view the last 20 lines of a file?
36. What does the `*` wildcard match?
37. What's the difference between `rm` and `rmdir`?

Chapter Quiz

1. To create a directory with parents:

- a) `mkdir dir`
- b) `mkdir -r dir`
- c) `mkdir -p dir` ✓ Correct
- d) `mkdir -m dir`

2. To copy a directory recursively:

- a) cp dir /dest
- b) cp -r dir /dest ✓ Correct
- c) cp -d dir /dest
- d) copy dir /dest

3. file? matches:

- a) file only
- b) file1, file2, file3 ✓ Correct
- c) file.txt
- d) All files starting with file

4. To force delete without prompts:

- a) rm -i
 - b) rm -v
 - c) rm -f ✓ Correct
 - d) rm -d
-

Chapter 6: User and Group Administration

Learning Objectives

After completing this chapter, you will be able to:

- Understand Linux user and group concepts
- Create and manage user accounts
- Create and manage groups
- Configure password policies
- Use user information commands

6.1 Understanding Users and Groups

User Types

Type	UID Range	Purpose
Root	0	Superuser with full access
System	1-999	Services and daemons
Normal	1000+	Regular users

```
$ id root
uid=0(root) gid=0(root) groups=0(root)

$ id mostafa
uid=1000(mostafa) gid=1000(mostafa) groups=1000(mostafa),10(wheel)
```

Groups

- **Primary Group:** Default group for new files
- **Secondary Groups:** Additional group memberships

6.2 User Configuration Files

/etc/passwd - User Information

Format: username:x:UID:GID:comment:home:shell

```
$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
mostafa:x:1000:1000:Mostafa Elkhoully:/home/mostafa:/bin/bash
```

Field	Description
username	Login name
x	Password placeholder
UID	User ID number
GID	Primary group ID
comment	User description (GECOS)
home	Home directory path
shell	Default shell

/etc/shadow - Password Information

Format: username:password:lastchange:min:max:warn:inactive:expire:

```
$ sudo cat /etc/shadow
root:$6$abc...xyz:18645:0:99999:7:::
mostafa:$6$def...uvw:18645:0:99999:7:::
```

Field	Description
password	Encrypted password (! = locked)
lastchange	Days since Jan 1, 1970
min	Days before change allowed
max	Days until change required
warn	Warning days before expiry
inactive	Days after expiry until disabled
expire	Account expiration date

6.3 Group Configuration Files

/etc/group

Format: groupname:x:GID:members

```
$ cat /etc/group
root:x:0:
wheel:x:10:mostafa,admin
developers:x:1001:mostafa,ahmed
```

/etc/gshadow

Contains group passwords and administrators.

6.4 Creating Users

useradd - Create User

Syntax: useradd [options] username

Option	Description
-m	Create home directory
-d	Specify home directory
-s	Specify shell
-g	Primary group
-G	Secondary groups
-c	Comment/description
-u	Specify UID

```
# Create user with defaults
$ sudo useradd ahmed

# Create user with options
$ sudo useradd -m -s /bin/bash -c "Ahmed Mohamed" ahmed

# Create user in specific group
$ sudo useradd -g developers -G wheel newuser
```

passwd - Set Password

Syntax: passwd [username]

```
$ sudo passwd ahmed
New password:
Retype new password:
passwd: password updated successfully

# Change your own password
$ passwd
```

View Defaults

```
$ useradd -D
GROUP=100
HOME=/home
INACTIVE=-1
EXPIRE=
SHELL=/bin/bash
SKEL=/etc/skel
```

6.5 Modifying Users

usermod - Modify User

Syntax: usermod [options] username

Option	Description
-l	Change login name
-L	Lock account
-U	Unlock account
-g	Change primary group
-G	Set secondary groups
-aG	Add to secondary group
-d	Change home directory
-s	Change shell

```
# Change primary group
$ sudo usermod -g developers ahmed

# Add to secondary group (append)
$ sudo usermod -aG wheel ahmed

# Lock user account
$ sudo usermod -L ahmed

# Unlock user account
$ sudo usermod -U ahmed
```

6.6 Deleting Users

userdel - Delete User

Syntax: userdel [options] username

```
# Delete user only
$ sudo userdel ahmed

# Delete user and home directory
$ sudo userdel -r ahmed
```

6.7 Group Management

groupadd - Create Group

```
$ sudo groupadd developers
$ sudo groupadd -g 2000 dbadmins
```

groupmod - Modify Group

```
$ sudo groupmod -n newname oldname
```

groupdel - Delete Group

```
$ sudo groupdel developers
```

gpasswd - Group Administration

```
# Add user to group
$ sudo gpasswd -a ahmed developers

# Remove user from group
$ sudo gpasswd -d ahmed developers

# Set group password
$ sudo gpasswd developers
```

6.8 Password Policies

chage - Change Age

Syntax: chage [options] username

Option	Description
-l	List aging info
-m	Min days between changes
-M	Max days between changes
-W	Warning days
-E	Expiration date

```
$ sudo chage -l ahmed
Last password change          : Feb 09, 2026
Password expires              : never
Account expires               : never

$ sudo chage -M 90 -W 7 ahmed
# Password expires in 90 days, warn 7 days before
```

6.9 User Information Commands

Command	Description
id	Show user/group IDs
whoami	Current username
who	Logged in users
w	Detailed user info
finger	User details
groups	Current user's groups

```
$ id ahmed
uid=1001(ahmed) gid=1001(ahmed) groups=1001(ahmed),10(wheel)

$ whoami
mostafa

$ who
mostafa pts/0      2026-02-09 12:30 (:0)

$ groups mostafa
mostafa : mostafa wheel developers
```

Practical Lab 6: User Management

Lab Objectives

- Create users and groups
- Manage passwords and policies
- Modify user properties

Lab Steps

Step 1: Create a group

```
$ sudo groupadd students
$ cat /etc/group | grep students
```

Step 2: Create users

```
$ sudo useradd -m -g students -c "Student One" student1
$ sudo useradd -m -g students -c "Student Two" student2
$ cat /etc/passwd | grep student
```

Step 3: Set passwords

```
$ sudo passwd student1
$ sudo passwd student2
```

Step 4: Add secondary group

```
$ sudo groupadd projects
$ sudo usermod -aG projects student1
$ id student1
```

Step 5: Check password aging

```
$ sudo chage -l student1
$ sudo chage -M 30 -W 5 student1
$ sudo chage -l student1
```

Step 6: Cleanup

```
$ sudo userdel -r student1
$ sudo userdel -r student2
$ sudo groupdel students
$ sudo groupdel projects
```

Review Questions

38. What file contains user passwords?
 39. What UID range is for normal users?
 40. How do you add a user to a secondary group?
 41. What command shows password aging info?
 42. What option deletes a user's home directory?
-

Chapter Quiz

1. Normal user UIDs start at:

- a) 0
- b) 100
- c) 500
- d) 1000 ✓ Correct

2. To add user to group without removing existing groups:

- a) usermod -G group user
- b) usermod -aG group user ✓ Correct
- c) groupadd user group
- d) addgroup user group

3. User passwords are stored in:

- a) /etc/passwd
- b) /etc/shadow ✓ Correct
- c) /etc/group
- d) /etc/users

4. To delete user with home directory:

- a) userdel user
 - b) userdel -r user ✓ Correct
 - c) rmuser user
 - d) deluser user
-

Page 43

File Type Indicators

Symbol	Type
-	Regular file
d	Directory
l	Symbolic link
b	Block device
c	Character device

7.3 Changing Permissions - Symbolic Mode

chmod Command (Symbolic)

Syntax: chmod [who][operator][permissions] file

Who	Operator	Permission
u (user)	+ (add)	r (read)
g (group)	- (remove)	w (write)
o (other)	= (set exactly)	x (execute)
a (all)		

Examples

```
# Add execute to owner
$ chmod u+x script.sh

# Remove write from group
$ chmod g-w file.txt

# Add read to others
$ chmod o+r file.txt

# Multiple modifications
$ chmod u+x,g-w,o-r file.txt

# Set exact permissions
$ chmod u=rwx,g=rx,o=r file.txt

# Add execute to all
$ chmod +x script.sh
$ chmod a+x script.sh

# Remove all permissions from others
$ chmod o-rwx private.txt

# Recursive (for directories)
$ chmod -R g+w Documents/
```

7.4 Changing Permissions - Octal Mode

Each permission has a numeric value:

- Read (r) = 4
- Write (w) = 2
- Execute (x) = 1

Add values for each category:

Permission	Calculation	Octal
rwX	4+2+1	7
rw-	4+2+0	6
r-X	4+0+1	5
r--	4+0+0	4
-wX	0+2+1	3
-w-	0+2+0	2
--X	0+0+1	1
---	0+0+0	0

Common Permission Sets

Octal	Symbolic	Typical Use
755	rxrx-rx-r-x	Executable files
644	rw-r--r--	Regular files
700	rwX-----	Private directories
600	rw-----	Private files
777	rxrxrxrxrx	Full access (unsafe!)

Examples

```
# rxrx-rx-r-x
$ chmod 755 script.sh

# rw-r--r--
$ chmod 644 document.txt

# rwX-----
$ chmod 700 private_dir/

# rw-----
$ chmod 600 secret.txt

$ ls -l script.sh
-rwxr-xr-x. 1 mostafa mostafa 100 Feb 9 12:00 script.sh
```

7.5 Changing Ownership

chown - Change Owner

Syntax: chown [options] owner[:group] file

```
# Change owner
$ sudo chown ahmed file.txt

# Change owner and group
$ sudo chown ahmed:developers file.txt

# Change group only
$ sudo chown :developers file.txt

# Recursive
$ sudo chown -R ahmed:developers project/
```

chgrp - Change Group

Syntax: chgrp [options] group file

```
$ sudo chgrp developers file.txt
$ sudo chgrp -R developers project/
```

7.6 Default Permissions (umask)

The **umask** determines default permissions for new files.

How umask Works

Files	Directories	
Base	666	777
Subtract umask	-022	-022
Result	644	755

Common umask Values

umask	File Result	Directory Result
022	644 (rw-r--r--)	755 (rwxr-xr-x)
027	640 (rw-r-----)	750 (rwxr-x---
077	600 (rw-----)	700 (rwx-----)

Using umask

```
$ umask
0022

$ umask 027
$ touch newfile
$ ls -l newfile
-rw-r-----. 1 mostafa mostafa 0 Feb 9 12:00 newfile
```

Practical Lab 7: Managing Permissions

Lab Objectives

- Create files with specific permissions
- Change permissions using both methods
- Modify ownership

Lab Steps

Step 1: Create test environment

```
$ mkdir ~/lab7  
$ cd ~/lab7  
$ touch file1.txt file2.txt script.sh
```

Step 2: View current permissions

```
$ ls -l
```

Step 3: Change permissions (symbolic)

```
$ chmod u+x script.sh  
$ chmod go-r file1.txt  
$ ls -l
```

Step 4: Change permissions (octal)

```
$ chmod 755 script.sh  
$ chmod 600 file2.txt  
$ ls -l
```

Step 5: Test umask

```
$ umask  
$ umask 077  
$ touch private.txt  
$ ls -l private.txt  
$ umask 022 # Reset
```

Step 6: Cleanup

```
$ cd ~  
$ rm -r lab7/
```

Review Questions

43. What does permission 755 mean?
 44. How do you add execute permission for owner?
 45. What does umask 022 result in for new files?
 46. What command changes file ownership?
 47. What is the difference between u, g, and o?
-

Chapter Quiz

1. Permission `rw-r-xr-x` in octal is:

- a) 644
- b) 755 ✓ Correct
- c) 777
- d) 700

2. To remove read from group:

- a) `chmod g-r file` ✓ Correct
- b) `chmod -g file`
- c) `chmod r-g file`
- d) `chmod g=r file`

3. `chown ahmed:devs file.txt` changes:

- a) Owner only
- b) Group only
- c) Both owner and group ✓ Correct
- d) Permissions

4. Default file permissions with umask 022:

- a) 777
 - b) 755
 - c) 644 ✓ Correct
 - d) 600
-

Chapter 8: Vi/Vim Text Editor

Learning Objectives

After completing this chapter, you will be able to:

- Understand Vi editor modes
 - Create and edit files with Vi
 - Navigate within files efficiently
 - Search and replace text
 - Customize Vi sessions
-

8.1 Introduction to Vi

Vi (Visual Editor) is the default text editor in Unix/Linux systems.

Why Learn Vi?

- Available on ALL Unix/Linux systems
- Essential for system administration
- Works without GUI
- Fast and powerful once mastered
- Required in emergency/recovery modes

Vi vs Vim

- **Vi**: Original visual editor
 - **Vim**: Vi Improved (more features)
 - Syntax highlighting
 - Mouse support
 - Multiple undo levels
-

8.2 Vi Modes

Vi operates in three modes:

Mode	Purpose	How to Enter
Command	Navigate, delete, copy	ESC
Insert	Type text	i, a, o, I, A, O
Last Line	Save, quit, search	:(colon)

```
+-----+
|  COMMAND MODE  |  <-- Default (ESC to enter)
+-----+-----+
|
|  i, a, o
v
+-----+-----+
|  INSERT MODE  |  <-- For typing text
+-----+-----+
|
|  ESC
v
+-----+-----+
|  LAST LINE MODE |  <-- For commands (:)
```

8.3 Starting Vi

Command	Description
vi	Open empty buffer
vi filename	Open/create file
vi +n filename	Open at line n
vi -r filename	Recover file
view filename	Read-only mode

```
$ vi document.txt
$ vi +10 document.txt    # Start at line 10
$ view /etc/passwd      # Read only
```

8.4 Entering Insert Mode

From Command mode, use these keys:

Key	Action
i	Insert before cursor
a	Append after cursor
I	Insert at line beginning
A	Append at line end
o	Open new line below
O	Open new line above

Press **ESC** to return to Command mode.

8.5 Navigation (Command Mode)

Character Movement

Key	Movement
h or ←	Left one character
l or →	Right one character
j or ↓	Down one line
k or ↑	Up one line

Word Movement

Key	Movement
w	Forward one word
b	Backward one word
e	End of current word

Line Movement

Key	Movement
O	Beginning of line
\$	End of line
^	First non-blank character

Screen Movement

Key	Movement
Ctrl+F	Page forward
Ctrl+B	Page backward
Ctrl+D	Half page down
Ctrl+U	Half page up

File Movement

Key	Movement
G	Last line of file
gg	First line of file
nG	Go to line n
:n	Go to line n

8.6 Editing (Command Mode)

Deleting Text

Key	Action
x	Delete character at cursor
X	Delete character before cursor
dw	Delete word
dd	Delete entire line
D	Delete to end of line
d\$	Delete to end of line
d0	Delete to beginning of line
5dd	Delete 5 lines

Changing Text

Key	Action
s	Substitute character
cw	Change word
cc	Change entire line
C	Change to end of line
r	Replace single character
R	Replace mode (overwrite)

Copy and Paste

Key	Action
yy	Yank (copy) line
yw	Yank word
5yy	Yank 5 lines
p	Paste below cursor
P	Paste above cursor

Undo and Redo

Key	Action
u	Undo last change
U	Undo all changes on line
Ctrl+R	Redo (in Vim)

8.7 Search and Replace

Searching

Command	Action
/pattern	Search forward
?pattern	Search backward
n	Next occurrence
N	Previous occurrence

/error	# Search for "error"
n	# Next match
N	# Previous match

Search and Replace

Syntax: :%s/old/new/options

Command	Action
:s/old/new/	Replace first on current line
:s/old/new/g	Replace all on current line
:%s/old/new/g	Replace all in file
:%s/old/new/gc	Replace all with confirmation

8.8 Saving and Quitting

Command	Action
:w	Save file
:w filename	Save as new file
:q	Quit (if no changes)
:q!	Quit without saving
:wq	Save and quit
:x	Save and quit
ZZ	Save and quit (no colon)

8.9 Vi Settings

Command	Action
:set nu	Show line numbers
:set nonu	Hide line numbers
:set ic	Ignore case in search
:set noic	Case sensitive search
:set showmode	Display current mode
:set syntax=on	Syntax highlighting

Practical Lab 8: Vi Editor Practice

Lab Objectives

- Create and edit files in Vi
- Practice navigation and editing
- Use search and replace

Lab Steps

Step 1: Create a file

```
$ vi practice.txt
```

Step 2: Enter insert mode and type

```
Press i, then type:  
This is line one.  
This is line two.  
This is line three.  
Press ESC
```

Step 3: Save the file

```
:w
```

Step 4: Practice navigation

```
gg      - Go to first line  
G       - Go to last line  
0       - Beginning of line  
$       - End of line
```

Step 5: Practice editing

```
dd      - Delete a line  
yy      - Yank a line  
p       - Paste below  
u       - Undo
```

Step 6: Search and replace

```
/line   - Search for "line"  
n       - Next match  
:%s/line/row/g - Replace all
```

Step 7: Save and quit

```
:wq
```

Vi Quick Reference Card

MOVEMENT		EDITING	
h j k l	- arrows	i	- insert
w b e	- words	a	- append
0 \$	- line	x	- delete char
gg G	- file	dd	- delete line
Ctrl+F/B	- page	yy	- yank line
SAVING		SEARCH	
:w	- save	/text	- forward
:q	- quit	?text	- backward
:wq	- both	n N	- next/prev
:q!	- force	:%s/a/b/g	replace

Chapter Quiz

1. Default mode when opening Vi:

- a) Insert
- b) Command ✓ Correct
- c) Last Line
- d) Visual

2. To insert text before cursor:

- a) a
- b) i ✓ Correct
- c) o
- d) I

3. To delete entire line:

- a) d
- b) x
- c) dd ✓ Correct
- d) D

4. To save and quit:

- a) :q
- b) :w
- c) :wq ✓ Correct
- d) :e

Chapter 9: Environment Variables and Shell Configuration

Learning Objectives

After completing this chapter, you will be able to:

- Understand environment variables
- View and set variables
- Customize shell configuration
- Create command aliases
- Use command history effectively

9.1 Understanding Environment Variables

Environment variables store system and user configuration values.

Common Environment Variables

Variable	Description	Example
\$HOME	User's home directory	/home/mostafa
\$USER	Current username	mostafa
\$SHELL	Default shell	/bin/bash
\$PATH	Command search path	/usr/bin:/bin
\$PWD	Current directory	/home/mostafa
\$HOSTNAME	Computer name	server1
\$LANG	Language setting	en_US.UTF-8
\$TERM	Terminal type	xterm-256color

9.2 Viewing Variables

View Single Variable

```
$ echo $HOME
/home/mostafa

$ echo $PATH
/usr/local/bin:/usr/bin:/bin

$ echo $USER
mostafa
```

View All Variables

```
# Environment variables
$ env
$ printenv

# All variables (including shell)
$ set
```

9.3 Setting Variables

Temporary Variables (Current Session)

```
# Set variable
$ MYVAR="Hello World"
$ echo $MYVAR
Hello World

# Export for child processes
$ export MYVAR="Hello World"
```

Permanent Variables

Add to configuration files for persistence.

9.4 Configuration Files

Order of Execution (Login Shell)

48. `/etc/profile` - Global settings for all users
49. `~/.bash_profile` or `~/.profile` - User login settings
50. `~/.bashrc` - User shell settings

File Descriptions

File	Scope	When Executed
<code>/etc/profile</code>	All users	Login shell
<code>/etc/bashrc</code>	All users	Every shell
<code>~/.profile</code>	User	Login shell
<code>~/.bash_profile</code>	User	Login shell
<code>~/.bashrc</code>	User	Every shell

Example ~/.bashrc

```
# User aliases
alias ll='ls -la'
alias la='ls -A'

# Custom PATH
export PATH=$PATH:$HOME/bin

# Custom prompt
PS1='[\u@\h \W]\$ '
```

Apply Changes

```
$ source ~/.bashrc
# or
$ . ~/.bashrc
```

9.5 Command Aliases

Aliases are shortcuts for commands.

Create Alias

```
$ alias ll='ls -la'
$ alias update='sudo yum update'
$ alias ..='cd ..'
```

View Aliases

```
$ alias
alias ll='ls -la'
alias ls='ls --color=auto'
```

Remove Alias

```
$ unalias ll
```

Bypass Alias

```
$ \ls          # Use backslash
$ /bin/ls      # Use full path
```

Permanent Aliases

Add to ~/.bashrc:

```
alias ll='ls -la'
alias cls='clear'
alias h='history'
```

9.6 Command History

Bash stores command history in ~/.bash_history

View History

```
$ history
 1  ls
 2  cd Documents
 3  cat file.txt
```

History Shortcuts

Shortcut	Action
↑ / ↓	Browse history
!!	Repeat last command
!n	Run command number n
!-n	Run nth previous command
!string	Run last command starting with string
^old^new	Replace in last command

Examples

```
$ ls /var/log
$ !!                # Runs: ls /var/log

$ !15               # Run command #15

$ !cat              # Last command starting with "cat"

$ cat file.txt
$ ^fle^file         # Runs: cat file.txt
```

History Settings

```
# In ~/.bashrc
HISTSIZE=1000      # Commands in memory
HISTFILESIZE=2000  # Commands in file
HISTCONTROL=ignoredups # Ignore duplicates
```

9.7 The PATH Variable

PATH defines where the shell searches for commands.

View PATH

```
$ echo $PATH  
/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin
```

Add to PATH

```
# Temporarily  
$ export PATH=$PATH:/home/mostafa/scripts  
  
# Permanently (add to ~/.bashrc)  
export PATH=$PATH:$HOME/bin:$HOME/scripts
```

Practical Lab 9: Shell Configuration

Lab Objectives

- Create custom aliases
- Modify environment variables
- Customize the shell

Lab Steps

Step 1: View current environment

```
$ echo $HOME  
$ echo $PATH  
$ echo $SHELL
```

Step 2: Create temporary variable

```
$ GREETING="Hello Linux"  
$ echo $GREETING  
$ export GREETING
```

Step 3: Create aliases

```
$ alias ll='ls -la'  
$ alias cls='clear'  
$ ll  
$ cls
```

Step 4: Edit ~/.bashrc

```
$ vi ~/.bashrc
# Add these lines:
alias myip='ip addr show'
export EDITOR=vi
# Save and exit
```

Step 5: Apply changes

```
$ source ~/.bashrc
$ myip
```

Step 6: Practice history

```
$ history | tail -5
$ !!
$ !-2
```

Review Questions

51. What variable contains your home directory path?
 52. Where should permanent aliases be placed?
 53. How do you apply changes to ~/.bashrc?
 54. What does !! do in the shell?
 55. How do you add a directory to PATH?
-

Chapter Quiz

1. To view the PATH variable:

- a) echo PATH
- b) echo \$PATH ✓ Correct
- c) print PATH
- d) show PATH

2. Permanent aliases should be in:

- a) /etc/password
- b) ~/.bashrc ✓ Correct
- c) /tmp/aliases
- d) ~/.bash_history

3. To repeat the last command:

- a) !-1
- b) !! ✓ Correct
- c) !last
- d) repeat

4. To apply .bashrc changes:

- a) restart
 - b) source ~/.bashrc ✓ Correct
 - c) reload bash
 - d) exit
-

Chapter 10: System Administration Basics

Learning Objectives

After completing this chapter, you will be able to:

- Switch between user accounts
 - Use sudo for administrative tasks
 - Monitor logged-in users
 - Safely shutdown and reboot systems
 - Get help using Linux documentation
-

10.1 Switching User Accounts

su - Switch User

Syntax: su [-] [username]

Command	Action
su username	Switch to user (keep environment)
su - username	Switch with user's environment
su - or su - root	Switch to root

```
$ whoami
mostafa

$ su - ahmed
Password:
$ whoami
ahmed

$ su -
Password:
# whoami
root
```

Run Single Command as User

```
$ su - ahmed -c "whoami"
ahmed
```

10.2 Using sudo

sudo allows permitted users to run commands as root.

Basic Usage

```
$ sudo command
$ sudo yum update
$ sudo useradd newuser
```

sudo vs su

Feature	su	sudo
Password	Root's password	Your password
Logging	Limited	Full audit log
Granularity	All or nothing	Per-command control
Security	Shares root password	More secure

Common sudo Commands

```
$ sudo -i      # Interactive root shell
$ sudo -s      # Shell as root
$ sudo -u user cmd # Run as specific user
$ sudo -l      # List allowed commands
```

The sudoers File

Located at `/etc/sudoers`. Edit with `visudo`:

```
$ sudo visudo
```

Example entries:

```
# User privilege specification
root    ALL=(ALL:ALL) ALL
mostafa ALL=(ALL:ALL) ALL

# Group sudo access
%wheel  ALL=(ALL)    ALL

# Allow specific commands
ahmed   ALL=(ALL) /usr/bin/yum, /usr/bin/systemctl
```

10.3 Monitoring Users

who - Who is Logged In

```
$ who
mostafa pts/0      2026-02-09 12:30 (:0)
ahmed    pts/1      2026-02-09 13:00 (192.168.1.10)
```

w - Detailed User Info

```
$ w
14:00:00 up 2 days, 3:30, 2 users, load average: 0.01, 0.02, 0.00
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
mostafa   pts/0    :0               12:30    0.00s  0.05s  0.01s  w
ahmed     pts/1    192.168.1.10    13:00    5:00   0.01s  0.01s  bash
```

Other User Commands

Command	Description
whoami	Current effective user
id	User and group IDs
users	List logged in users
finger user	User details
last	Login history

10.4 System Shutdown and Reboot

shutdown Command

Syntax: shutdown [options] time [message]

Command	Action
shutdown -h now	Halt immediately
shutdown -r now	Reboot immediately
shutdown -h +10	Halt in 10 minutes
shutdown -r 22:00	Reboot at 10 PM
shutdown -c	Cancel scheduled shutdown
shutdown -k now	Send warning only

Quick Commands

Command	Action
poweroff	Power off system
reboot	Reboot system
halt	Halt system
init 0	Shutdown (runlevel 0)
init 6	Reboot (runlevel 6)
systemctl poweroff	Power off
systemctl reboot	Reboot

```
# Immediate reboot
$ sudo reboot

# Immediate shutdown
$ sudo poweroff

# Schedule shutdown with message
$ sudo shutdown -h +15 "System maintenance"
```

10.5 Linux Documentation

man - Manual Pages

Syntax: man [section] command

```
$ man ls
$ man 5 passwd    # Section 5 (file formats)
```

Manual Sections

Section	Content
1	User commands
2	System calls
3	Library functions
4	Device files
5	File formats
6	Games
7	Miscellaneous
8	Admin commands

Navigating man Pages

Key	Action
Space	Next page
b	Previous page
/text	Search forward
n	Next search match
q	Quit

Other Help Commands

```
$ man -k keyword    # Search manuals
$ whatis command    # One-line description
$ apropos keyword    # Search descriptions
$ command --help     # Built-in help
$ info command       # Detailed documentation
```

Practical Lab 10: System Administration

Lab Objectives

- Practice user switching
- Use sudo for administration
- Safely shutdown the system

Lab Steps

Step 1: Check current user

```
$ whoami  
$ id
```

Step 2: View logged in users

```
$ who  
$ w  
$ users
```

Step 3: Practice sudo

```
$ sudo -l  
$ sudo cat /etc/shadow | head -3
```

Step 4: Get help on commands

```
$ man passwd  
$ whatis useradd  
$ ls --help | head -10
```

Step 5: Test shutdown (cancel immediately!)

```
$ sudo shutdown -h +5 "Test shutdown"  
$ sudo shutdown -c # Cancel!
```

Troubleshooting Exercises

56. User cannot login:

- Check if account is locked: `passwd -S username`
- Check if shell is valid: `grep username /etc/passwd`

57. Permission denied on sudo:

- Verify user is in wheel group: `groups username`
- Check sudoers: `sudo visudo`

58. **Command not found:**

- Check PATH: `echo $PATH`
 - Find command location: `which command`
-

Review Questions

- 59. What's the difference between `su` and `sudo`?
 - 60. What file controls sudo access?
 - 61. What command shows who is logged in with details?
 - 62. How do you cancel a scheduled shutdown?
 - 63. What are the man page sections?
-

Chapter Quiz

1. To switch to root with root's environment:

- a) `su root`
- b) `su -` ✓ **Correct**
- c) `sudo root`
- d) `switch root`

2. sudo configuration is in:

- a) `/etc/passwd`
- b) `/etc/sudoers` ✓ **Correct**
- c) `/etc/sudo`
- d) `~/.sudo`

3. To cancel scheduled shutdown:

- a) `shutdown -stop`
- b) `shutdown -c` ✓ **Correct**
- c) `shutdown -cancel`
- d) `shutdown -x`

4. Manual page section for file formats:

- a) 1
 - b) 3
 - c) 5 ✓ **Correct**
 - d) 8
-

Appendix A: Command Quick Reference

Category	Commands
Navigation	cd, pwd, ls
Files	touch, mkdir, cp, mv, rm
Viewing	cat, more, less, head, tail
Users	useradd, userdel, passwd, usermod
Groups	groupadd, groupdel, groupmod
Permissions	chmod, chown, chgrp, umask
Editor	vi, vim, nano
System	shutdown, reboot, poweroff
Help	man, whatis, --help

Appendix B: Important Files

File	Purpose
/etc/passwd	User accounts
/etc/shadow	User passwords
/etc/group	Groups
/etc/sudoers	Sudo configuration
/etc/profile	Global shell settings
~/.bashrc	User shell settings
~/.bash_history	Command history

Congratulations!

You have completed the Operating Systems Practical Training Book. Continue practicing these skills to become proficient in Linux system administration.